NHSAST
المدرسة الوطنية العليا لتكنولوجيا الأنظمة المستقلة
NATIONAL HIGHER SCHOOL OF AUTONOMOUS SYSTEMS TECHNOLOGY

Machine Learning

Lecture 01
Introduction to Machine Learning

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OVERVIEW

Today's **Agenda**

1. **Rule-Based Systems vs ML**
   Comparison, differences, adaptability, and real-world example.
2. **Learning paradigms**
   Supervised vs Unsupervised
3. **CRISP-DM Methodology**
   Phases of the data mining process, objectives, and workflow structure.
4. **Model Selection Concepts**
   Train–test split, validation strategies and generalization.

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ML vs Rule-based Systems

- *What rule-based systems are*
- *Why they fail for complex problems*
- *Comparison with ML*

Why Do We Need Machine Learning

- Motivation :
  - Spam detection
  - Recommendation systems
  - Fraud detection
  - Image and speech recognition

➢ Can we realistically write rules for all these problems?

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Traditional Programming : Rule-based System

- What is a rule-based system ?
  - Logic explicitly written by humans
  - IF–THEN rules
  - Deterministic behavior

- Example

```
IF email contains "free money" → spam
ELSE → not spam
```

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Limitations of Rule-Based Systems

- Rules become complex and explode in number
- Difficult to maintain
- Fail when data changes
- Hard to capture uncertainty
- Too rigid

  - Some problems are **too complex** for manual rules !

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Machine Learning Systems

- **Learns patterns from data** : identifies relationships and trends that we might not explicitly program
- **Improves with experience** : gets better as it sees more examples
- **Adapts to new situations** : handles changes in the environment or variations in input

  - Adaptivity and continous learning

Example : Decoding Handwritten Text

Rule-Based System

"If there is a vertical line with a small horizontal stroke near the middle, then it is the letter 'f'."

- Fixed Program with explicit rules
- Cannot handle variation in handwriting
- Only recognizes perfectly printed letters

Machine Learning System

- Learns from many examples
- Adapts to different handwriting styles
- Recognizes new ways an 'f' is written

Rule-Based vs Machine Learning

Rule-Based Systems	Machine Learning
Rules written by humans	Rules learned from data
No learning	Learns from experience
Hard to scale	Scales with data
Brittle	Adaptive

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What is Machine Learning ?

"The field of study that gives computers the ability to learn without being explicitly programmed."

— Arthur Samuel, 1959

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Learning Paradigms

- *How does ML work ?*
- *Supervised Learning*
- *Unsupervised Learning*

Types of Learning

- Learning : Broad domain with many subfields
- ML divided into different **learning paradigms**
- Show how tasks and methods are organized and related

Learning

Supervised **Unsupervised**

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Supervised Learning – Core Idea

- Build a model to **predict** an **answer** or **label** already provided

- During training :
  - Inputs **X** and corresponding labels **Y** are provided
  - The learning algorithm observes both **X** and **Y**
  - It learns a mapping **X → Y**

- **Goal**
  - Given a new input **X**, predict the most appropriate output **Y**

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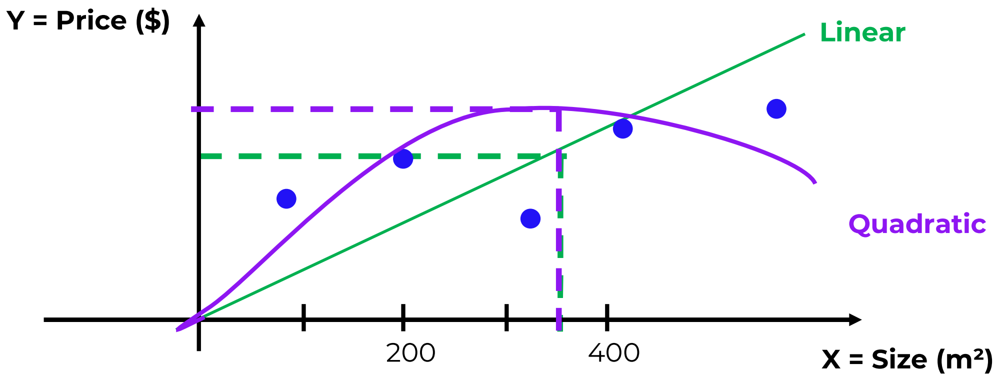
Supervised Learning – Regression Example

Predict the price of a house according to its size



Mapping X → Y

Regression problem : Value Y trying to predict is continuous

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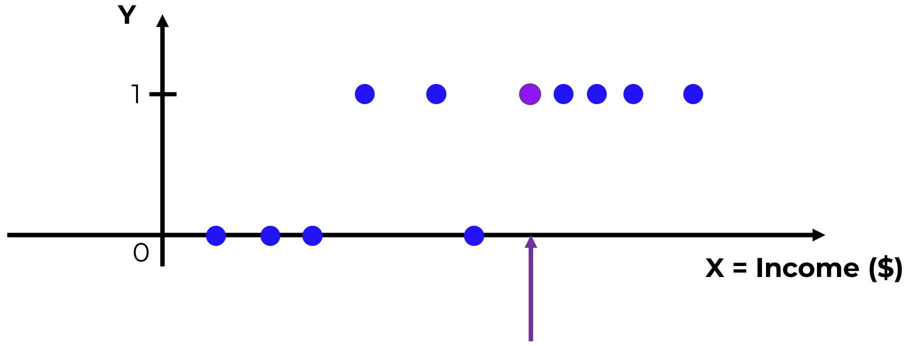
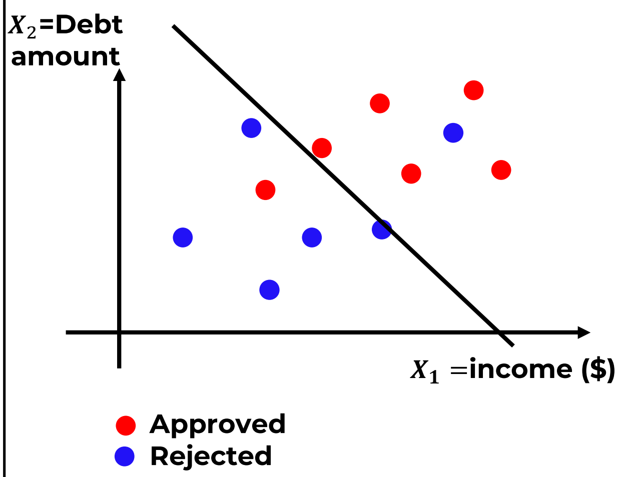
Supervised Learning – Classification Example

Classify loan approval knowing the income



Classification problem : Y takes a discrete number of variables

Supervised Learning – Classification



- Usually : **more features than just one**
- We'll see model that allows to fit data to separate both types

Prediction vs Classification – What's the Difference ?

- Classification is a type of prediction, But not all predictions are classifications.

Prediction

- Estimate an unknown output from data
- Output can be:
  - A number (regression)
  - A class label (classification)
  - A probability / score

Classification

- Output is a discrete label
- Often based on:
  - A score or probability
  - A decision threshold

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Unsupervised Learning – Core Idea

- Discover patterns or structure in data without **labeled output**

- During training :
  - Data has inputs **X** only
  - There is no correct answer **Y** provided

- **Goal**
  - The algorithm looks for **similarities, differences, or hidden patterns**
  - It **groups or organizes** data based on shared features

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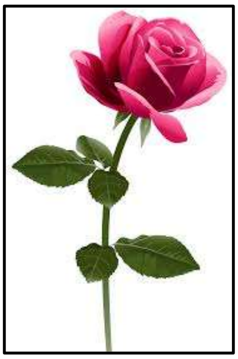
Unsupervised Learning – Clustering

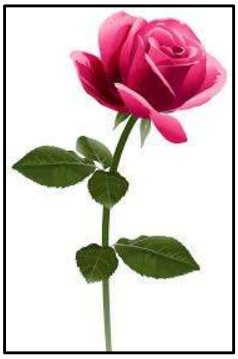
- Groups data points based on similarity




Unsupervised Learning – Clustering

- Groups data points based on similarity

Pink

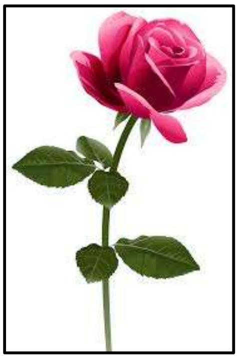



Yellow


Unsupervised Learning – Clustering

- Groups data points based on similarity

Short petals




Long petals


Unsupervised Learning – Association

- Finds rules that describe how features occur together

Example : Shopping basket

- Algorithm find patterns such as
  - Customers who buy bread often buy butter
  - Customers who buy diapers often buy wipes
- By observing customers’ shopping baskets, we can understand which items are frequently bought together
- Used for product placement and recommendations

Unsupervised Learning – Dimensionality Reduction

- Reduces the number of features while keeping important information.

Example : Student data

- Math score
- Science score
- English score
- History score
- Art score

- Reduced to :
  - Scientific skills
  - Literary and artistic subjects

Supervised vs Unsupervised

Learning

- **Supervised**
  - **Regression** – Predict a **continuous value**
  - **Classification** – Predict a **discrete label**
- **Unsupervised**
  - **Clustering** – **Discover structure** without label
  - **Association** – **Find patterns**
  - **Dim. Reduction** – **Compress/Visualize** data

Supervised Learning - RECAP

Rule-Based Systems	Machine Learning
Data	Labeled data (X,Y)
Goal	Predict known outputs
Learns from	Input-Output pairs
Typical tasks	Classification, Regression

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Unsupervised Learning - RECAP

Rule-Based Systems	Machine Learning
Data	Unlabeled data (X only)
Goal	Discovers patterns or structure
Learns from	Input data only
Typical tasks	Clustering, Association, Dimensionality Reduction

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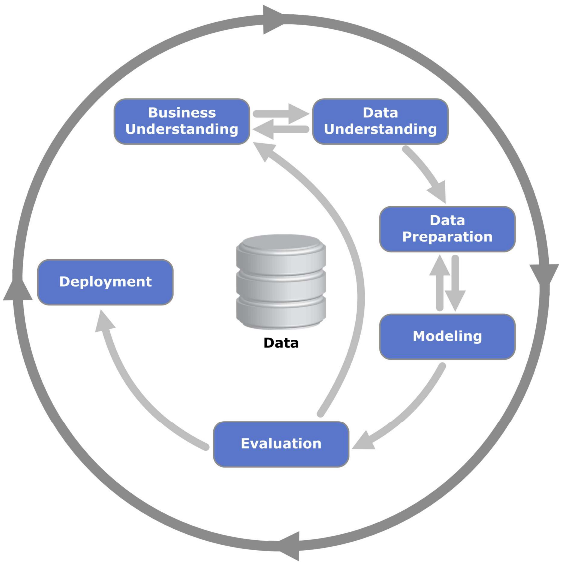
CRISP-DM Methodology

- *Data mining process*
- *Objective*
- *Workflow structure*

CRIPS-DM Methodology – How to organize projects ?



- CRIPS-DM – CRoss-Industry Standard Process for Data Mining
- Widely used, cyclical framework for structuring data science project
- 6 key phases

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1 – Business Understanding

1. Identifying & Defining the Business Problem

- Understand the issue and how it can be solved.
- Ask the right questions: *Do we really need Machine Learning?*

Example:
- Users complain about spam messages.
- Analyze the extent of the problem.
- Determine if ML can help.
- If not, propose an alternative solution.

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1 – Business Understanding

2. Define the goal : Make it **clear** and **measurable**

Example:
- Reduce the number of spam messages.
- Reduce the number of complaints about spam.

Set a measurable target
- **Reduce** spam messages by **50%**

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2 – Data Understanding

1. Analyze Available Data Sources

- Determine what data is available and whether we need more.

Key Questions:

- What data sources exist?
- Is the data good enough and reliable?
- Is it tracked correctly?
- Is the dataset large enough?
- Do we need to collect more data?

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2 – Data Understanding

2. How data influences the goal

- Available data may affect the goal defined in Step 1.
- If data is insufficient or unreliable, go back and adjust the goal or approach.

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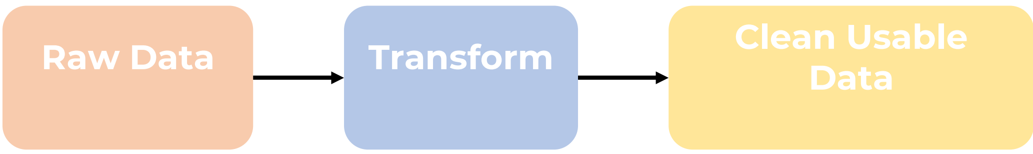
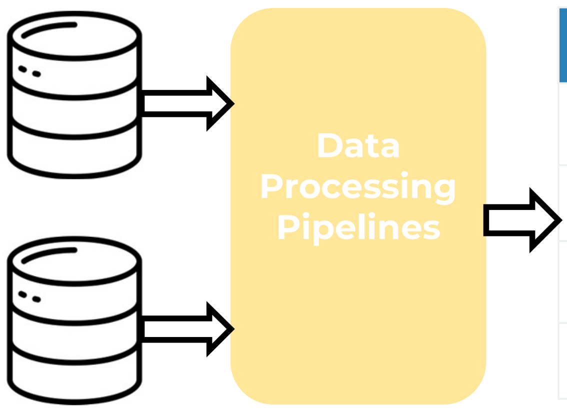
3 – Data Preparation

1. Transform the data so it can be put into an ML algorithm

- Clean the data, remove the noise
- Build the pipelines
- Convert into tabular form



3 – Data Preparation



Sender	Receiver	Subject	...	Spam
PrSAiyfa@nvrsgge.us	example@gmail.com	Congratulations you've....		TRUE
school@school.com	example@gmail.com	Your Transcript		FALSE
CAA @bigcorp.com	example@gmail.com	Free		TRUE
FED@enterprise.com	example@gmail.com	You've won a....		TRUE

4 – Modeling

1. This is where **the actual ML happens**

- Training a model:
  - Try different models
  - Select the best performing one
- Common model types (we'll cover in this course):
  - Logistic Regression
  - Decision Trees
  - Neural Networks
  - ...and more

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4 – Modeling

1. This is where **the actual ML happens**

- Sometimes you go back to data preparation to improve performance by
  - Adding more features
  - Fixing data issues

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5 – Evaluation

1. Measure how well the model solves the business problem, not just how well it performs mathematically.

Key questions:
- Is the model good enough?
- Have we reached the goal defined in Step 1?
- Do the evaluation metrics show a real improvement?

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5 – Evaluation

1. Measure how well the model solves the business problem, not just how well it performs mathematically.

Evaluation
- Has spam been reduced?
- By how much?
- Measure this **on the test set** (unseen data)

- A model with high accuracy is not necessarily successful if it does not meet the business target.

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5 – Evaluation

2. Analyze whether the goal and metrics were appropriate, then decide to iterate, deploy, or stop.

Retrospective questions

- Was the initial goal realistic and achievable?
- Did we measure the right thing?
- Do the results truly reflect real-world performance?

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5 – Evaluation

2. Analyze whether the goal and metrics were appropriate, then decide to iterate, deploy, or stop.

Based on the evaluation, we may decide to:

- Adjust or redefine the initial goal
- Improve the data or the model
- Deploy the model to more users (or all users)
- Stop the project if the benefit is insufficient

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5 – Deployment

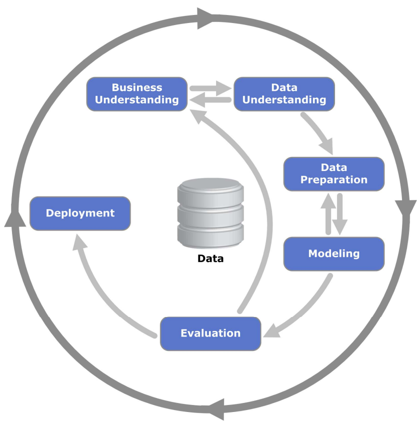
Evaluation and Deployment often come together

- Roll the model to all users
- Proper monitoring
- Ensuring the **quality** and **maintainability**

Learn and Iterate !

ML projects require many iterations !

- Reflect on the process and revisit the Business Understanding step
- Determine if the current solution is sufficient or needs improvement
- After some time, review results and start the cycle again if needed
- Redefine the business model, adjust the approach, or incorporate new data sources.



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Model Selection Concepts

- *Train–test split*
- *Validation strategies*
- *Generalization*

What Are We Trying to Achieve?

- Learn from data
- Make predictions on **new**, **unseen data**
- Performance on future data matters most

➢ We care about how the model behaves in the real world

- We train a model using data
- We need a way to measure its performance

Should we evaluate it on the same data it learned from?

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Exam Intutive Analogy

- Study material → learning phase (training the data)
- Final exam → evaluation phase (new data)

- Scenario : Exam uses the same questions as the study sheet
- Score : 100%

➢ High score, but no guarantee of real understanding as the score does not reflect real ability

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Need for Data Separation

- Evaluation must be **independent** of training
- Otherwise, the score is misleading
- We need **unseen** data to evaluate performance fairly

➢ We divide the data into different parts, each with a clear role.
  - Training set
  - Validation set
  - Test set

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Data Split Overview

Training | Validation | Test

Training set
- Used to learn patterns from data
- Model parameters are fitted here
- Only data the model learns from

Validation set
- Compare different models
- Choose model settings

Test set
- Used only once
- Represents future, unseen data
- Provides the final performance estimate
- Must remain untouched